

Yun Seok Kim

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RESEARCH INTERESTS

Hardware–software co-design for emerging ML applications and deployment of efficient AI system at scale.

EDUCATION

Yonsei University <i>M.S. in Electrical and Electronic Engineering (GPA: 3.96/4.3)</i>	Seoul, Republic of Korea <i>Expected Feb. 2027</i>
Sungkyunkwan University <i>B.S. in Electronic and Electrical Engineering (GPA: 3.97/4.5)</i>	Suwon, Republic of Korea <i>Feb. 2025</i>

RESEARCH POSITIONS

Yonsei University	Seoul, Republic of Korea
Graduate Researcher, <i>Embedded Systems and Computer Architecture Lab</i> (Advisor: Prof. Won Woo Ro)	<i>Mar. 2025 – Present</i>
Sungkyunkwan University	Suwon, Republic of Korea
- Undergraduate Researcher, <i>Computer Architecture & Systems Lab</i> (Advisor: Prof. Seokin Hong) - Undergraduate Researcher, <i>V-Lab</i> (Advisor: Prof. Eunbyung Park)	<i>Jun. 2024 – Aug. 2024</i> <i>Dec. 2023 – Feb. 2024</i>

PUBLICATIONS

*Equal contribution.

Under Review

- Yun Seok Kim***, Sunbin Kim*, Mingu Jung, Enhyeok Jang, and Won Woo Ro. “Autobahn: Efficient Autoregressive Video Generation Based on Semantic-Temporal Reduction.” Under review at the *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2027.

Conference Papers

- Yun Seok Kim** and Won Woo Ro. “TAPAS: A Case Study on Topology-Aware Multi-PIM in Model Parallelism.” *International Conference on Consumer Electronics Asia (ICCE-Asia)*, Busan, Korea, 2025.
- Hoje Lee, **Yun Seok Kim**, and Won Woo Ro. “Real-Time Access Pattern-Based Page Migration Policy in CXL Memory Systems.” *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Jeju, Korea, 2025.

ONGOING PROJECTS

Memory-Centric Architecture Using Reconfigurable PIM Devices	<i>Mar. 2025–Present</i>
- Analyzed Flash Attention and investigated efficient execution methods using reconfigurable processing-in-memory architectures. - Explored hardware/software co-design techniques, including workload mapping, data allocation, and coordination among hardware components.	

TEACHING EXPERIENCE

Teaching Assistant, Yonsei University

- EEE 4473: Embedded System Lab. (Undergraduate) *Spring 2025*
 - Contributed lectures on programmable logic and processing systems using Zynq 7020 kit
 - Organized practice questions and had Q&A sessions during lecture

TECHNICAL SKILLS

- **Frameworks:** CUDA, Nsight System/Compute, PyTorch
- **Programming Languages:** C/C++, Python, Verilog

SCHOLARSHIPS AND AWARDS

Fellowships and Scholarships

- BK (Brain Korea) 21 Research Fellowship, *Yonsei University* 2025-2026
- Graduate Student Research Assistant (GSRA), *Yonsei University* 2025

Awards

- Outstanding Presentation Award, *Automotive System IC Fusion Human Resource Research Center (ASRC)* 2026

CONTRIBUTED TALKS

Adadvanced LLM Systems: Agentic AI

- Large Language Model (LLMs) Seminar, *Yonsei University* Feb. 2026

Understanding Multi-GPU Programming

- Advanced GPU Seminar, *Yonsei University* Aug. 2025

Advanced Branch Prediction and Dynamic Scheduling

- Adadvanced Computer Architecture Seminar, *Yonsei University* Feb. 2025

EXTRACURRICULAR ACTIVITIES

Professional Affiliations

- Graduate Student Member, *Institute of Electrical and Electronics Engineers (IEEE)* 2025-Present
- Trainee, *Automotive System IC Fusion Human Resource Research Center (ASRC)* 2025-Present

Military Service

- Battalion Commander Interpreter (KATUSA), *1-38 FAB, 2nd Infantry Division, 8th US Army* 2020-2022

LANGUAGE PROFICIENCY

Korean(Native), English(TOEFL: 5.0)